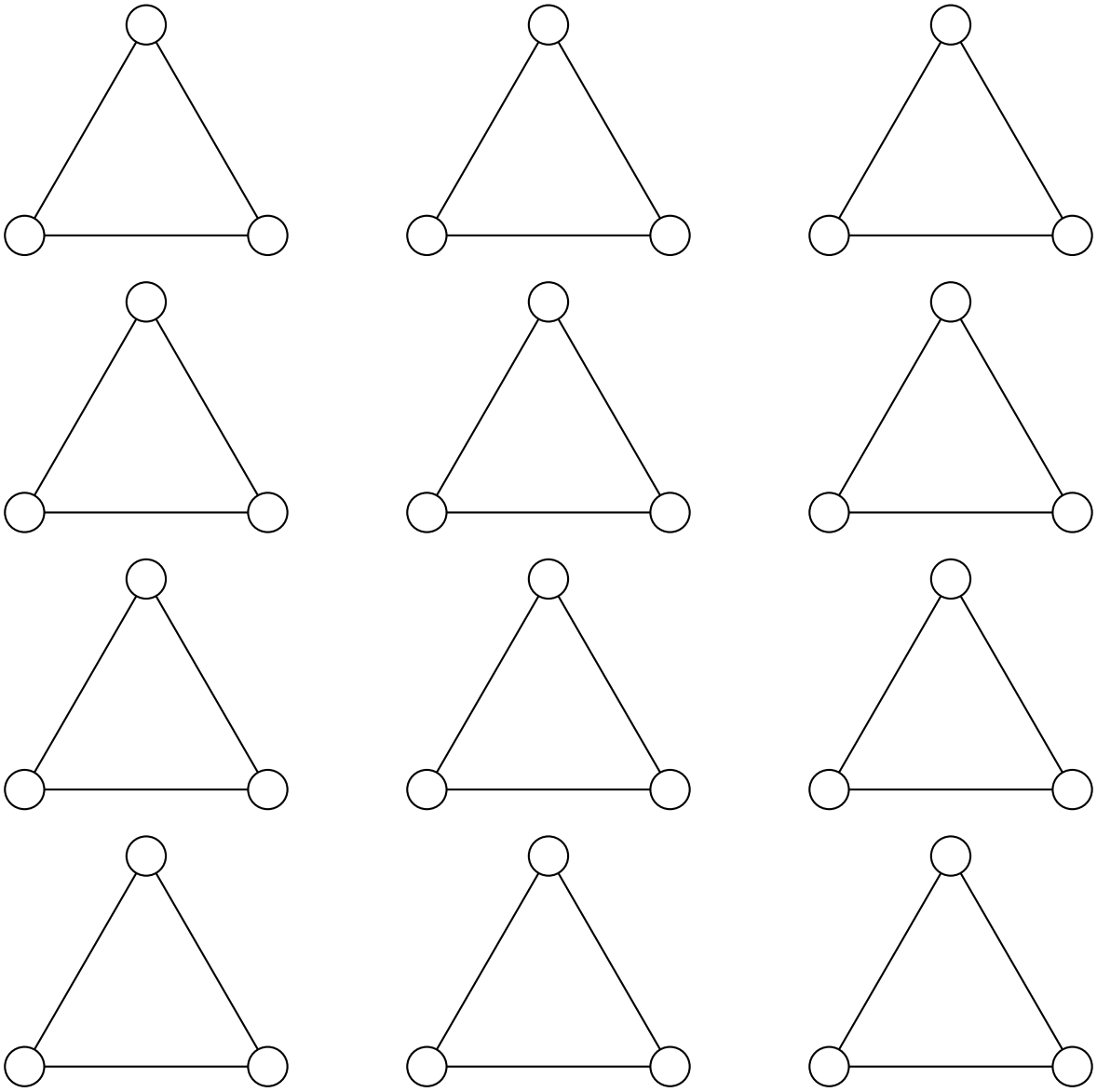
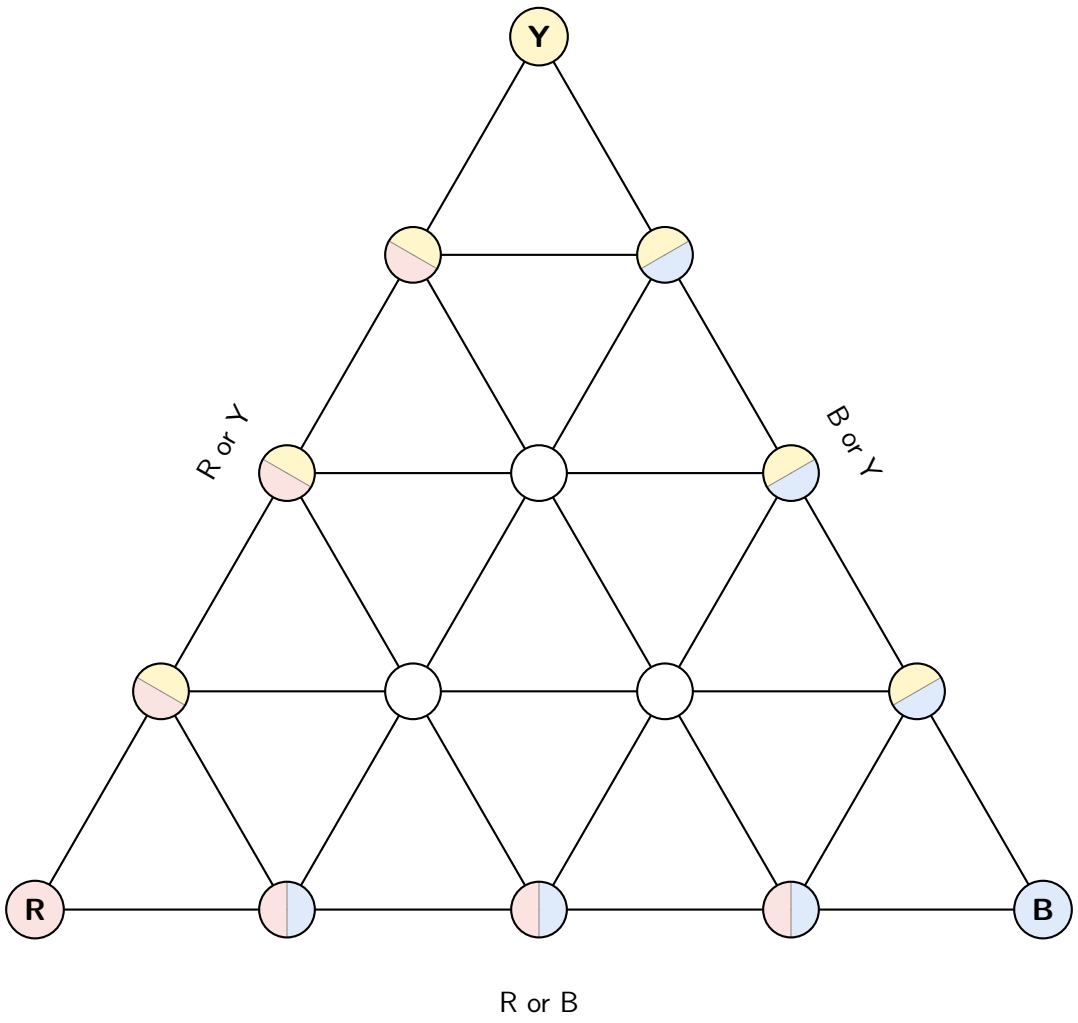
Put R, B, or Y on every empty dot. Keep the printed letters. Each outside side allows only the letters beside it. A little triangle has no lines inside.

Problem 3: Fill these triangles with R, B, and Y in different ways, and mark the doors in each. Which fillings have exactly one door? Could a triangle have three doors?



Problem 5: Fill the dots so that exactly three little triangles have R, B, and Y. Can all the routes that start at doors on the bottom side end outside?



Problem 6: Can the dots be filled so that no little triangle has R, B, and Y? Give a reason that covers every filling allowed by the side rules.

